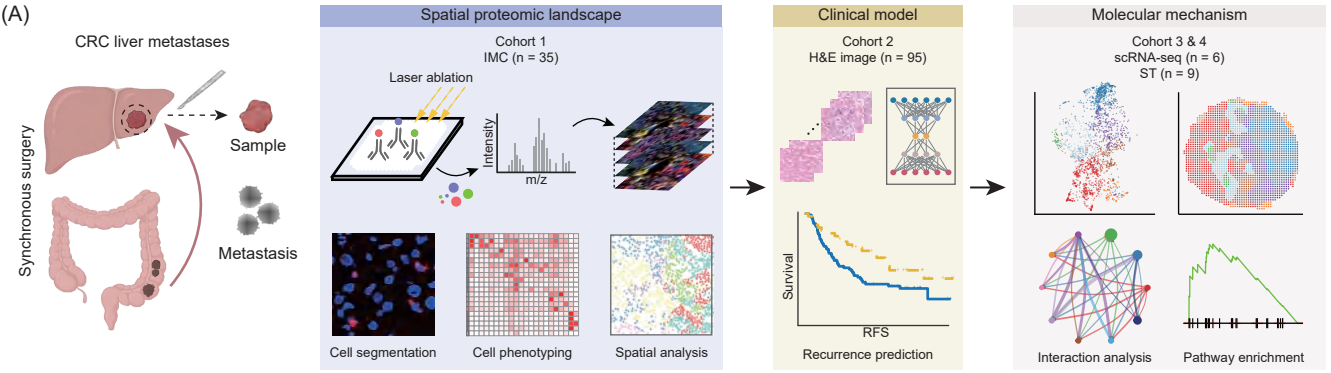
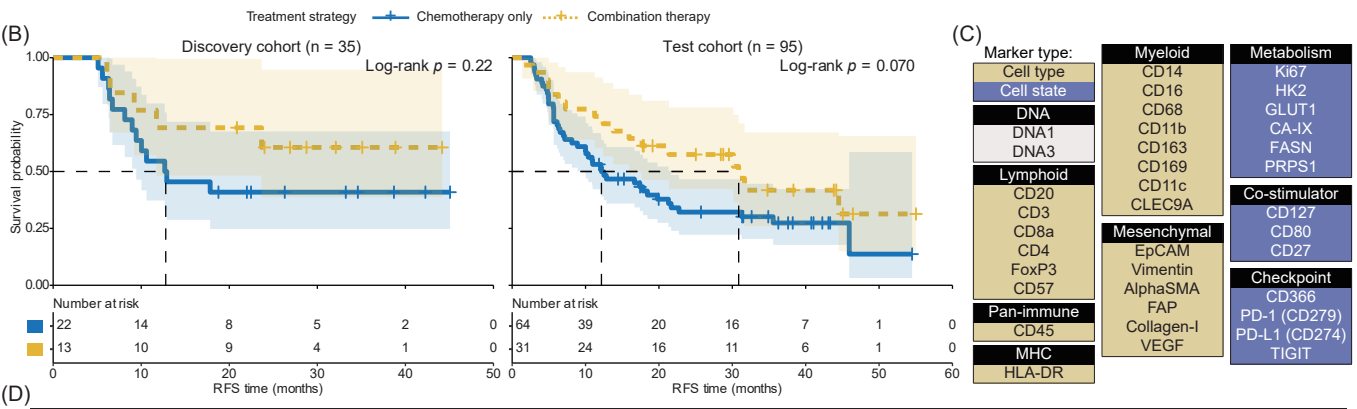
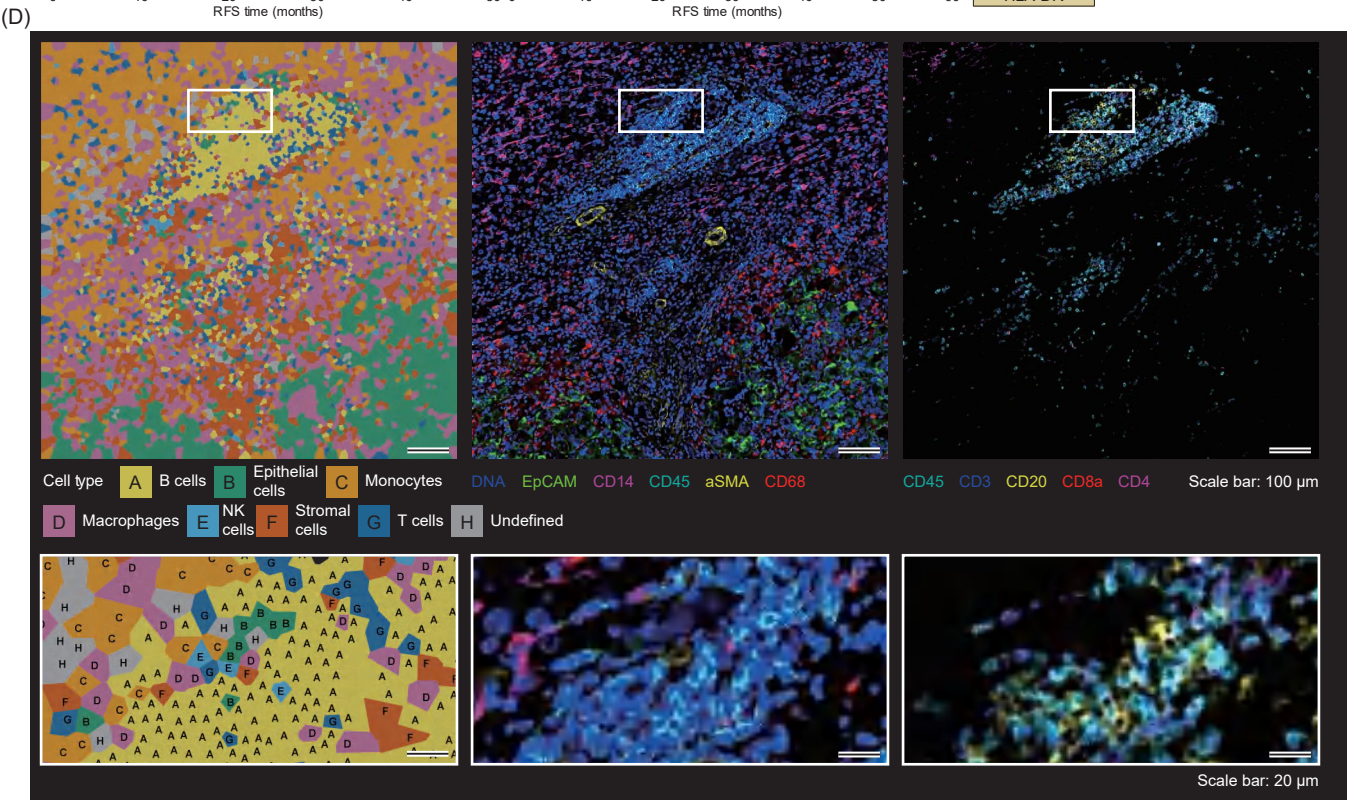

(A)
CRC liver metastases
Synchronous surgery
Sample
Metastasis
Spatial proteomic landscape
Cohort 1
IMC (n = 35)
Laser ablation
Intensity
m/z
Cell segmentation
Cell phenotyping
Spatial analysis
Clinical model
Cohort 2
H&E image (n = 95)
Survival
RFS
Recurrence prediction
Molecular mechanism
Cohort 3 & 4
scRNA-seq (n = 6)
ST (n = 9)
Interaction analysis
Pathway enrichment


Treatment strategy
Chemotherapy only
Combination therapy
(B)
Discovery cohort (n = 35)
Log-rank p = 0.22
Test cohort (n = 95)
Log-rank p = 0.070
Survival probability
Number at risk
RFS time (months)
(C)
Marker type:
Cell type
Cell state
DNA
DNA1
DNA3
Lymphoid
CD20
CD3
CD8a
CD4
FoxP3
CD57
Pan-immune
CD45
MHC
HLA-DR
Myeloid
CD14
CD16
CD68
CD11b
CD163
CD169
CD11c
CLEC9A
Mesenchymal
EpCAM
Vimentin
AlphaSMA
FAP
Collagen-I
VEGF
Metabolism
Ki67
HK2
GLUT1
CA-IX
FASN
PRPS1
Co-stimulator
CD127
CD80
CD27
Checkpoint
CD366
PD-1 (CD279)
PD-L1 (CD274)
TIGIT


(D)
Cell type
A B cells
B Epithelial cells
C Monocytes
D Macrophages
E NK cells
F Stromal cells
G T cells
H Undefined
DNA EpCAM CD14 CD45 aSMA CD68
CD45 CD3 CD20 CD8a CD4
Scale bar: 100 μm
Scale bar: 20 μm